

Enhanced LLR for Soft-Decision FEC in Amplified Spatial Division Multiplexing IMDD Systems

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Abstract—We propose an enhanced log-likelihood ratio (E-LLR) technique for soft-decision forward error correction (SD-FEC) in intensity-modulation direct-detection (IMDD) systems with optical amplifiers and demonstrate its effectiveness in 7-core weakly coupled fiber transmission. In amplified IMDD, square-law photodetection and modulator nonlinearities cause amplified spontaneous emission noise to produce non-Gaussian signal statistics, invalidating conventional LLR assumptions and degrading SD-FEC performance. To address this issue, we abandon the classic additive white Gaussian noise (AWGN) model and introduce a channel model based on multi-parameter chi-square distributions that more accurately reflect optical channel noise. Unlike previous Gaussian and chi-square approaches, E-LLR uses a training sequence to estimate the variance and non-centrality parameters for each pulse-amplitude modulation (PAM) level, generating M distinct noncentral chi-square distributions for precise LLR computation in FEC decoding. Experimental results show that the proposed approach improves the receiver sensitivity by 1.13dB and reduces the decoding iterations of FEC compared with AWGN-model based schemes. These improvements contribute to reduced system complexity and decoding latency, while still achieving the required bit error rate threshold.

Index Terms—Log-likelihood ratio, soft-decision forward error correction, intensity modulation and direct detection, optical amplifier, spatial division multiplexing.

I. INTRODUCTION

THE surging demands of emerging applications such as artificial intelligence and cloud computing, e.g., Ope-

nAI, DeepSeek, have posed unprecedented challenges to the data transmission rates and bandwidth capacities of global communication networks [1]. Traditional single-core fiber communication systems have greatly enhanced spectral efficiency by adopting high-order modulation formats and wavelength division multiplexing (WDM), boosting per-core throughput from hundreds of Gb/s to several Tb/s [2], [3], [4]. However, as modulation orders and transmission rates continue to rise, the system's signal-to-noise ratio (SNR) requirements escalate sharply, and nonlinear impairments, such as the Kerr effect within the fiber core, and inter-channel crosstalk in WDM systems become increasingly pronounced, pushing conventional single-core fibers toward the Shannon limit. To overcome these bottlenecks, space-division multiplexing (SDM) based on multi-core fibers (MCFs) has emerged, offering a promising solution to substantially increase the capacity of optical communication systems, particularly for data center interconnects [5].

For short-reach SDM systems, intensity modulation–direct detection (IMDD) has emerged as the preferred solution for efficient optical communication owing to its simple architecture and reduced system complexity. However, the insertion loss from fan-in/fan-out devices combined with an increasing baud rate has sharply escalated the optical power demand on the SDM systems [6]. Insufficient optical power can cause a significant drop in the signal-to-noise ratio (SNR), thereby deteriorating detection performance. Consequently, optical amplifiers have become indispensable components, as they not only compensate for the power attenuation caused by fiber transmission and device insertion loss but also boost the signal strength at the receiver, ensuring the stable and reliable operation of systems at high data rates [7]. Although optical amplifiers boost the optical power at the receiver, they also introduce amplified spontaneous emission (ASE) noise, which presents a challenge for signal detection. Therefore, the integration of forward error correction (FEC) codes is essential to ensure link reliability in amplified SDM systems [8].

Unlike hard-decision FEC (HD-FEC) which relies solely on binary symbol decisions, soft-decision FEC (SD-FEC) conveys per-symbol reliability metrics, yielding markedly superior error-correction performance and coding gain. Therefore, to guarantee the performance benefits of soft-decision FEC, it is essential to compute the log-likelihood ratios (LLRs) fed into the decoder with high precision. However, most existing SD-

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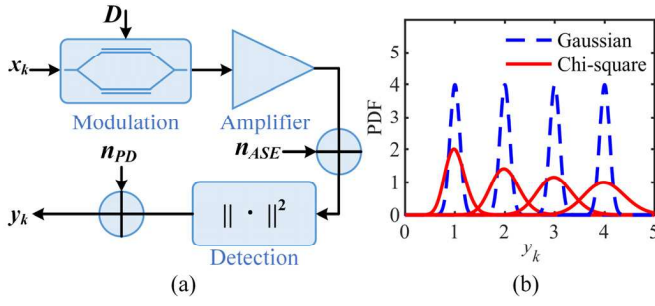


Fig. 1. (a) Channel model of an IMDD system with an optical amplifier, (b) PDF of PAM-4 after square-law detection.

FEC schemes adopt LLR formulas derived from an additive white Gaussian noise (AWGN) channel assumption, which only accounts for flat Gaussian noise impairments [9]. In SDM IMDD systems equipped with cascaded optical amplifiers, the accumulation of amplifier noise, modulator nonlinearity, and beat noise generated by the photodetector's square-law detection cannot be accurately characterized by a simple AWGN-based channel model. Previous studies have employed chi-square (CS) distributions to model optical channels, yet reference [10] only analyzes the channel capacity without addressing improvements in SD-FEC. Reference [11] evaluates convolutional codes, Turbo codes under an ideal CS channel in simulation, demonstrating advantages over the symmetric Gaussian model, but does not account for the impact of modulator nonlinearities and other practical noise effects on the CS distribution parameters.

To mitigate device nonlinearity and beat noise while enhancing decoding performance, this letter pioneers the introduction and application of an enhanced LLR estimation scheme based on a multi-parameter CS (MPCS) distribution in a 20-km weakly-coupled 7-core fiber pulse-amplitude modulation (PAM) transmission system employing optical amplifiers, integrating it into a SD-FEC decoder to substantially optimize decoding performance. The experimental results indicate that, compared to the conventional LLR model preferred in coherent detection, the received optical power sensitivity is improved by 1.13 dB by using the proposed E-LLR. Moreover, the number of iterations required in the low-density parity-check (LDPC) code can be reduced, thereby decreasing the system complexity and latency.

II. PRINCIPLES OF E-LLR CALCULATION

A. Channel Model

Fig. 1(a) depicts the channel model for an IMDD system with an optical amplifier, where the ASE noise constitutes the primary noise source. In the discrete-time model, the input electrical signal at time k , x_k , and the output of the PD, y_k , have the relation of

$$y_k = |D + x_k + n_{ASE}|^2 + n_{PD} \quad (1)$$

Here, D is the DC bias of the modulator. The n_{ASE} and n_{PD} respectively represent the ASE noise of the amplifier and the noise of the PD. Since the optical ASE is the predominant noise, the noise impact brought by PD is ignored to simplify

the model [12]. Therefore, the conditional probability density function (PDF) for the IMDD channel with optical amplifiers follows a non-central CS distribution with two degrees of freedom instead of a traditional Gaussian distribution in coherent detection [9] and its PDF is expressed as:

$$p(y_k|x_k) = \frac{1}{2\sigma^2} \exp\left(-\frac{y_k + \mu}{2\sigma^2}\right) I_0\left(\frac{\sqrt{y_k\mu}}{\sigma^2}\right) \quad (2)$$

where $\mu = (D + x_k)^2$ is the noncentrality parameter. $x_k = A_1, A_2, \dots, A_M$, where $A_i, i = 1, 2, \dots, M$, is the symbol amplitude value of M -ary PAM. σ^2 is noise variance. $I_0(\cdot)$ denotes the modified Bessel function of the first kind. Fig. 1(b) shows a schematic diagram of the PDF $p(y_k|x_k)$ for the Gaussian and CS distributions when $\mu = 1, 2, 3, 4$ and $\sigma^2 = 0.01$. It can be observed that the Gaussian distribution imposes a uniform noise influence across all amplitude levels, whereas the CS distribution exhibits amplitude-dependent variability.

B. Principle of E-LLR

In the SD-FEC decoding scheme, the first step is to extract input information from the entire data block of the detected signal. This includes the computation of prior LLRs. For each input symbol y_k at the decoder input, the prior log-likelihood ratio is defined as:

$$LLR(b_m) = \ln\left(\frac{p(b_m = 1|y_k)}{p(b_m = 0|y_k)}\right) \quad (3)$$

here, b_m denotes the m -th coded bit mapped onto the M -ary PAM symbol at the transmitter, $m = 1, 2, \dots, \log_2 M$. In most FEC studies, a common approach is to model the channel as an AWGN channel. In this case, the received signal can be written as: $y_k = x_k + n$ where n is an additive white Gaussian noise variable with a variance of σ^2 , and its PDF is given by:

$$P_{AWGN}(n) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{n^2}{2\sigma^2}\right) \quad (4)$$

However, it previously was noted that in SDM IMDD systems with an optical amplifier, the square-law detection causes the PDF of the received signal to exhibit a CS distribution. Furthermore, factors such as modulator nonlinearity cause deviations in the non-central parameters and noise variance for each PAM amplitude. Consequently, estimating the LLRs using traditional Gaussian and CS distributions based on a noise variance and standard constellation point would yield inaccurate results.

To achieve accurate Log-Likelihood Ratio (LLR) estimation under actual transmission conditions, the proposed E-LLR estimation method utilizes training sequences to evaluate the non-central parameters of the received sequence and the independent variance corresponding to each non-central parameter. For example, the non-central parameter and noise variance at each amplitude level of the M -ary PAM constellation $A_i, i = 1, 2, \dots, M$, can be estimated by:

$$\begin{aligned} \mu_i &= \text{mean}(T_i(A_i)) \\ \sigma_i^2 &= \text{mean}[(y_i(A_i) - T_i(A_i))^2] \end{aligned} \quad (5)$$

where $y_i(A_i)$ and $T_i(A_i)$ denote the received symbols and its mean for amplitude level A_i in the training symbols,

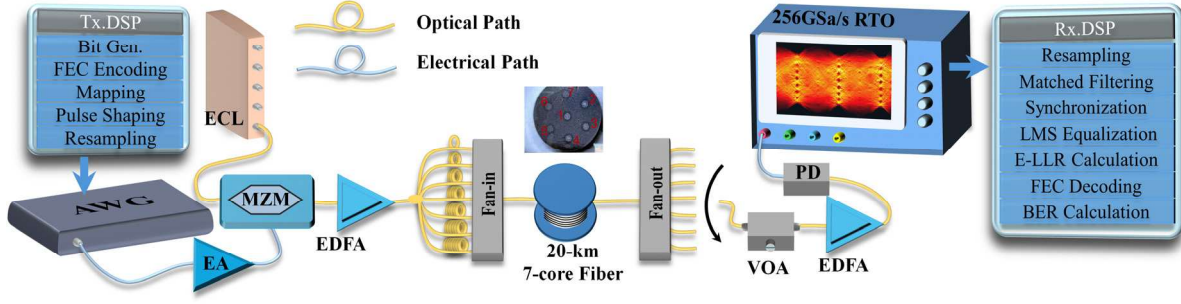


Fig. 2. Experimental setup.

respectively. Compared with the traditional CS distribution, the MPCS estimated based on training sequences can obtain M groups of accurate non-central parameters and variances, enabling the SD-FEC decoder to tolerate modulator nonlinearity and channel noise. Therefore, Eq. (2) can be rewritten as:

$$p(y_k|x_k) = \frac{1}{2\sigma_i^2} \exp\left(-\frac{y_k + \mu_i}{2\sigma_i^2}\right) I_0\left(\frac{\sqrt{y_k}\mu_i}{\sigma_i^2}\right) \quad (6)$$

Subsequently, the received observations and the estimated noise variances across the M amplitudes are employed to compute the LLRs as:

$$\begin{aligned} LLR(b_m) &= \ln \left(\frac{\sum_{s \in S_{m,1}} \frac{1}{2\sigma_s^2} \exp\left(-\frac{y_k + \mu_s}{2\sigma_s^2}\right) I_0\left(\frac{\sqrt{y_k}\mu_s}{\sigma_s^2}\right)}{\sum_{s \in S_{m,0}} \frac{1}{2\sigma_s^2} \exp\left(-\frac{y_k + \mu_s}{2\sigma_s^2}\right) I_0\left(\frac{\sqrt{y_k}\mu_s}{\sigma_s^2}\right)} \right) \end{aligned} \quad (7)$$

here, $S_{m,1}$ and $S_{m,0}$ denote the sets of PAM amplitudes in which the m -th bit is 1 and 0, respectively. Finally, the computed E-LLR values are forwarded to the SD-FEC decoder.

III. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

Fig. 2 shows the experimental setup of a 20-Gbaud PAM-8 IMDD system over a 20-km 7-core weakly coupled MCF. At the transmitter, the digitally generated PAM-8 signal, which was processed through LDPC encoding, mapping, pulse shaping, and resampling, was subsequently loaded into an arbitrary waveform generator (AWG) with a sampling rate of 128 GSa/s. The LDPC code featured a length of 64,800 bits with a code rate of 5/6. The radio frequency (RF) output from the digital-to-analog converter (DAC) was boosted by an electrical amplifier (EA) before driving the modulator. Here, an external cavity laser (ECL) providing an optical carrier with a 100-kHz linewidth and a wavelength of 1550 nm was modulated by a commercial Mach-Zehnder intensity modulator (MZM) with a 40-GHz bandwidth, using the amplified RF signal. After modulation, the optical signals were further enhanced by an erbium-doped fiber amplifier (EDFA); thereafter, a 1:8 coupler created multiple signal replicas which were decorrelated via optical delay lines of different lengths. Finally, these signals were injected into the 20-km, weakly coupled, 7-core MCF using a fan-in assembly.

At the receiver, the signals from the 7-core fiber were detected individually. The received optical power (ROP) was

regulated using a variable optical attenuator (VOA). Due to the absence of a transimpedance amplifier after the PD that had a 3-dB bandwidth of 50 GHz, an EDFA was employed prior to the PD for signal amplification. The detected electrical signal was captured by a digital real-time oscilloscope (RTO) operating at 256 GSa/s. Subsequently, offline DSP was performed, which encompassed resampling, matched filtering, synchronization, least mean square (LMS) equalization, computation of the E-LLRs, LDPC decoding, and bit error rate (BER) measurement. In this work, it is assumed that the outer layer applies a hard decision threshold defined by KP4-FEC at 2.4×10^{-4} [13] for post-hard decision evaluation, thereby ensuring compatibility with current Ethernet standards while maintaining low power consumption, minimal latency, and cost-effectiveness.

B. Experimental Results

Fig. 3(a) illustrates the histograms of LLR for all output bits at a ROP of -11.5 dBm. In this figure, Gaussian (blue) and E-LLR (red) respectively denote the PDFs corresponding to the use of Eq. (4) and Eq. (6) in LLR calculation. It is seen that the histogram based on the proposed E-LLR shifts outward, which would lead to a lower post-FEC BER.

Fig. 3(b) presents the BERs for the Gaussian, CS and the E-LLR approach at various ROPs. The peak-to-peak voltage (Vpp) of the signal at the transmitter is 500 mV and the number of iterations of the LDPC is 5. Insets I and II show the eye diagram and the PDF of the received signal. Compared with the Gaussian approach, the E-LLR scheme enhances the sensitivity at the KP4-FEC threshold by 1.13 dB. The CS-based LLR scheme also provides improvement, but it exhibits 0.86-dB penalty compared to the proposed E-LLR.

Fig. 3(c) shows that the performance advantage of the proposed E-LLR is also maintained as the Vpp of the transmitted signal varies. When Vpp increases, modulation nonlinearity becomes prominent, and the performance degrades. However, the BER of E-LLR always outperforms other schemes at the same Vpp.

Fig. 3(d) illustrates the influence of the number of LDPC iterations on the BER. By providing more precise LLR values, the E-LLR scheme requires less ROP to reach the KP4-FEC threshold, regardless of the number of iterations. Fig. 3(e) shows the BER versus the number of iterations at the same ROP. The E-LLR scheme consistently outperforms other schemes for each fixed number of LDPC iterations. At a

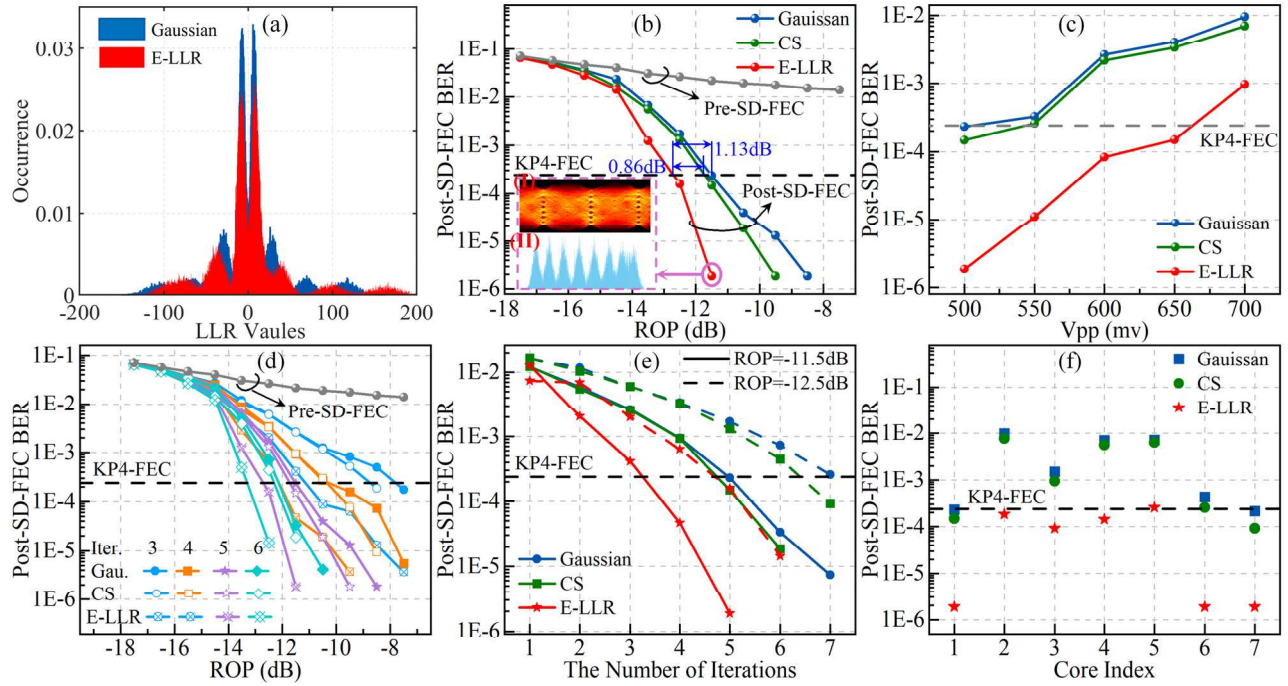


Fig. 3. (a) The histograms of LLR of all output bits, (b) the BER performance versus ROP, inset I is the eye diagram and II is the PDF of PAM-8, (c) Post-SD-FEC BER versus Vpp at a ROP of -11.5 dBm, (d) the BER performance versus ROP under different number of iterations, (e) the BER performance versus LDPC iterations at ROPs of -11.5 dBm and -12.5 dBm, (f) the BER performance of different cores at a ROP of -11.5 dBm.

fixed ROP, the E-LLR scheme requires fewer iterations to achieve the KP4-FEC threshold, thereby reducing the decoding complexity and latency.

Finally, Fig. 3(f) illustrates the BER across multiple cores of the 7-core fiber at a ROP of -11.5 dBm. Notably, the E-LLR scheme's post-LDPC BER is lower than that of the conventional calculation method in every core and remain consistently below the KP4-FEC threshold.

IV. CONCLUSION

We have proposed an E-LLR for soft-decision FEC in optically-amplified IMDD systems and experimentally demonstrated its performance advantage in an IMDD system over a 20-km, 7-core fiber with optical amplifiers. The proposed E-LLR is based on the MPCS distribution to combat non-typical Gaussian noise and nonlinear noise in IMDD systems. Experimental results demonstrate that, at the KP4-FEC threshold, compared with conventional Gaussian and CS distribution, the proposed E-LLR respectively improves the receiver sensitivity by 1.13 dB and 0.86 dB, and reduces the number of decoding iterations of LDPC. This performance advantage is also maintained for each core in the multi-core fiber transmission.

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